

Sky-Guard EO Tracking System

Visible-light and thermal imaging for target confirmation



Overview

An electro-optical tracking subsystem combining visible-light imaging, infrared thermal imaging, multi-spectral fusion, and servo tracking. It delivers day/night target detection, tracking, identification, and intelligent warning for low, slow, and small aerial targets.

Key Capabilities

- Visible-light + thermal imaging channels with multi-spectral data fusion.
- 48× optical continuous zoom with automatic focus and optical defogging.
- Horizontal 360° continuous rotation and -90° to +90° pitch coverage.
- Radar guidance, spectrum-positioning guidance, GPS, environmental data, and power-management access.

Technical Specifications

Visible Camera	1920 × 1080 @ 25/30 fps
Optical Lens	F6.5–312 mm, 48× continuous zoom
Thermal Imaging	Uncooled 640 × 512; 75 mm fixed lens
Day UAV Detection	≥ 2.5 km detection; ≥ 2 km tracking
Night UAV Detection	≥ 1.2 km detection; ≥ 1 km tracking
Gimbal	360° pan; -90° to +90° tilt; ±0.01° preset accuracy
Power / Weight	≤ 30 W steady-state; ≤ 8 kg
Protection	IP66

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